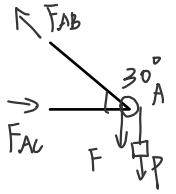
2-16　简易起重设备的计算简图如图2-2-23所示。已知斜杆AB用两根63mm×40mm×4mm不等边角钢组成，钢的许用应力[σ]＝170MPa。当提起重量为P＝15kN的重物时，试校核斜杆AB的强度？



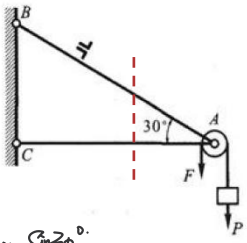


图2-2-23

由物体平衡：$F=P$.

对A点，$\begin{cases}\sum F_y=0 \Rightarrow 2F=F_{AB}\cdot\sin30^\circ \\ \sum F_x=0 \Rightarrow F_{AC}=F_{AB}\cdot\cos30^\circ\end{cases}$

$\therefore F_{AB}=4F$

查表得 $S=8.116\times10^{-4}m^2$

$\sigma_{AB}=\frac{F_{AB}}{S}=\frac{60\times10^3}{8.116\times10^{-4}}=7.4\times10^7Pa$

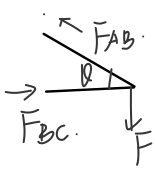
$=74MPa<170MPa=[\sigma]$

$\therefore$ AB安全

2-17　简单桁架及其受力如图2-2-25所示，水平杆BC的长度l保持不变，斜杆AB的长度可随夹角θ的变化而改变。两杆由同一材料制造，且材料的许用拉应力与许用压力相等。要求两杆内的应力同时达到许用应力，且结构的总重量为最小时，试求：

（1）两杆的夹角θ值；

（2）两杆横截面面积的比值。



$\begin{cases}\sum F_x=0 \Rightarrow F_{BC}=F_{AB}\cos\theta \\ \sum F_y=0 \Rightarrow F=F_{AB}\sin\theta\end{cases}$

$\begin{cases}F_{AB}=\frac{1}{\sin\theta}\cdot F \\ F_{BC}=\cot\theta\cdot F\end{cases}$

$\sigma_{AB}=\sigma_{BC}$

$\frac{F}{S_{AB}}\frac{1}{\sin\theta}=\cot\theta\frac{F}{S_{BC}}=[\sigma]$

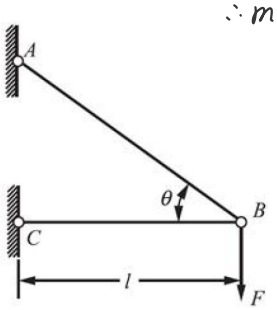


图2-2-25

$m=\rho V$

$\therefore m=\rho S_{AB}\frac{l}{\cos\theta}+\rho S_{BC}\cdot l$

$=\frac{\rho l F}{[\sigma]}\left(\frac{1}{\sin\theta\cos\theta}+\frac{\cos\theta}{\sin\theta}\right)$

$=\frac{\rho l F}{[\sigma]}\frac{2\cos^2\theta+\sin^2\theta}{\sin\theta\cos\theta}=\frac{\rho l F}{[\sigma]}\left(\tan\theta+\frac{2}{\tan\theta}\right)$

当 $\tan^2\theta=2$ 时，取极小值

此时 $\theta\approx54.74^\circ$　$\frac{S_{BC}}{S_{AB}}=\cos\theta=\frac{\sqrt{2}}{3}$

2-18　一桁架受力如图2-2-27所示。各杆都由两个等边角钢组成。已知材料的许用应力[σ]＝170MPa，试选择杆AC和CD的角钢型号。

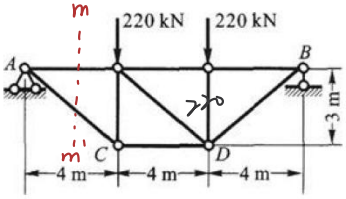


图2-2-27

对整体分析：

$\sum F=0$

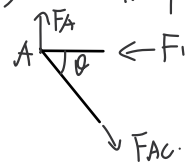
$\therefore F_A+F_B=440kN$

$\sum M_A=0$

$\therefore 220k\cdot4+220k\cdot8=12\cdot F_B$

$\therefore\begin{cases}F_A=220kN \\ F_B=220kN\end{cases}$

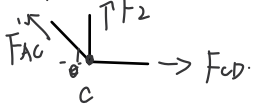
b) m-m' 截开



$\therefore F_{AC}\cdot\sin\theta=F_A$

$\therefore F_{AC}=\frac{1100}{3}kN$

对点C



$\therefore F_{CD}=F_{AC}'\cdot\cos\theta$

$=\frac{1100}{3}\times\frac{4}{5}kN$

$=\frac{880}{3}kN$

$\begin{cases}\frac{F_{AC}}{S_{AC}}\le[\sigma] \\ \frac{F_{CD}}{S_{CD}}\le[\sigma]\end{cases}$

$\therefore S_{AC}\ge21.57cm^2$

$S_{CD}\ge17.26cm^2$

查表可得 AC可选2根 80mm×7mm等边角钢

CD可选2根 75mm×6mm的等边角钢